

Diffusion and Flow-Based Models for Controllable Image Generation

Abstract

The landscape of generative modeling has undergone a paradigm shift with the advent of diffusion models and flow-based approaches, transitioning from adversarial training to likelihood-based and score-matching objectives. While early diffusion probabilistic models demonstrated remarkable sample quality, their reliance on stochastic differential equations (SDEs) and iterative denoising often resulted in high computational costs and limited controllability. Recently, Flow Matching (FM) and Rectified Flow models have emerged as powerful alternatives, offering deterministic ordinary differential equation (ODE) trajectories that are inherently straighter, faster to sample, and more amenable to precise control. This survey provides a comprehensive review of the state-of-the-art in diffusion and flow-based models specifically tailored for controllable image generation. We synthesize over 300 recent academic works to construct a detailed taxonomy of methodological advancements, ranging from theoretical unifications of probability transport to practical architectures for spatial, semantic, and attribute-level control. We examine core innovations including trajectory straightening techniques, condition-aware reparameterization, adapter-based control mechanisms, and training-free inference strategies. Furthermore, we explore the extension of these models to diverse domains such as medical imaging, remote sensing, video synthesis, and robotics. Finally, we discuss persistent challenges regarding evaluation bottlenecks, safety concerns, and the trade-offs between fidelity, diversity, and inference efficiency, outlining critical directions for future research.

1. Introduction

Generative modeling aims to learn the underlying probability distribution of data to synthesize novel, high-fidelity samples. Historically, Generative Adversarial Networks (GANs) dominated this field, yet they suffered from training instability and mode collapse. The emergence of Diffusion Probabilistic Models (DPMs) addressed these issues by framing generation as a reverse denoising process governed by Stochastic Differential Equations (SDEs) [1]. These models, including Denoising Diffusion Probabilistic Models (DDPM) and Score-Based Generative Models, achieved state-of-the-art results in image synthesis by iteratively removing noise from a Gaussian prior [1, 2]. However, the stochastic nature of the reverse SDE and the curved trajectories of the probability flow often necessitate hundreds or thousands of function evaluations (NFEs), hindering real-time applications and precise controllability [3].

In response, Flow Matching (FM) and related continuous normalizing flow (CNF) frameworks have gained prominence. FM formulates generative modeling as learning a vector field that transports a source distribution (typically Gaussian noise) to a target data distribution via an ODE [4, 5]. A key insight driving recent advancements is that the choice of the probability path between source and target significantly impacts sampling efficiency. By constructing straight-line paths, known as Rectified Flows, models can achieve high-quality generation with drastically fewer integration steps [6, 7, 8]. This deterministic nature not only accelerates inference but also simplifies the application of control signals, as the mapping from noise to data becomes a well-defined, invertible transformation [9, 10].

Controllability—the ability to guide generation using spatial layouts, semantic masks, textual attributes, or reference images—is paramount for practical deployment. Early control methods in diffusion models relied on classifier guidance or fine-tuning entire backbones, which was compu-

tationally prohibitive [11]. Recent works have introduced parameter-efficient adapters, such as ControlNet and T2I-Adapter, which inject conditions into frozen pre-trained models [12, 13]. In the realm of flow models, control is further enhanced by the ability to steer velocity fields directly, perform posterior sampling without retraining, and leverage the geometric properties of straight trajectories for editing and inversion [9, 14, 15].

This survey synthesizes the rapid progress in diffusion and flow-based controllable generation. We organize the literature into coherent methodological families, covering theoretical foundations, architectural innovations for conditioning, acceleration techniques, and domain-specific applications. We highlight how the transition from stochastic diffusion to deterministic flow matching has unlocked new capabilities in controllability, enabling everything from fine-grained attribute manipulation [16] to complex multi-modal scene composition [17, 18]. By analyzing over 300 distinct studies, we aim to provide a rigorous technical overview of the field, identifying trends, comparing methodological efficacy, and pinpointing open challenges.

2. Method

The methodological landscape of controllable diffusion and flow-based generation is vast, encompassing theoretical reformulations of probability transport, novel architectural designs for condition injection, and sophisticated inference-time steering mechanisms. We categorize these advancements into several key directions: theoretical unification and trajectory optimization, architectural strategies for conditional modeling, training-free and posterior sampling techniques, multi-modal and compositional control, and domain-specific adaptations.

2.1 Theoretical Foundations and Trajectory Optimization

The core distinction between traditional diffusion models and modern flow-based approaches lies in the formulation of the generative trajectory. Understanding and optimizing this trajectory is fundamental to achieving both speed and controllability.

2.1.1 Unification of Diffusion and Flow Matching Recent theoretical work has established that diffusion models and flow matching are instances of a unified framework based on learning time-dependent vector fields. The “Unified Transport Framework” posits that both paradigms learn vector fields inducing marginal families governed by continuity or Fokker-Planck equations [19]. Specifically, Flow Matching regresses a neural network vector field onto target conditional vector fields that generate fixed Gaussian probability paths, bypassing the intractable marginal integrals required in score matching [4]. This simulation-free training allows for unbiased gradients and faster convergence compared to standard diffusion objectives [4].

Theoretical analyses have further linked these models through the lens of Stochastic Optimal Control (SOC). It has been proven that Flow Matching is a special case of Diffusion Bridge models where the drift coefficient approaches zero [20]. While Diffusion Bridges incorporate an active drift term yielding lower SOC costs and more stable trajectories for inverse problems, Flow Matching relies on linear interpolation coefficients, which can violate optimal transport principles in complex distributions [20]. Furthermore, the “Generator Matching” framework extends these concepts to arbitrary Continuous-Time Markov Processes (CTMPs), unifying discrete and continuous state spaces under a single regression objective on analytically derived conditional paths [21].

2.1.2 Trajectory Straightening and Rectification A primary motivation for adopting flow-based models is the ability to “straighten” the probability flow, reducing the number of integration steps required for high-fidelity generation. Rectified Flow models learn neural velocity fields that solve ODEs transporting source distributions to targets along straight paths, reducing sampling steps from thousands to merely ten or fewer [22, 6].

Several techniques have been proposed to enforce this straightness. “Learning Straight Flows by Learning Curved Interpolants” formulates straight flow learning as a bi-level optimization problem, minimizing vector field curvature subject to Conditional Flow Matching constraints [6]. Similarly, “Sequential Flow Straightening” (SEQRF) segments the ODE time domain into intervals to bound global truncation error, training on segmented pairs to reduce curvature before distilling the model [23]. The “Balanced Conic Rectified Flow” method integrates real data inversions into reflow steps via Spherical Linear Interpolation (Slerp), anchoring ODE paths to the true data manifold to mitigate distributional drift [8].

Advanced straightening methods also explore the geometry of the transport map. “Designing Optimal Transport Flows” replaces the standard Gaussian prior with low-frequency image projections, proving that identity coupling between data and low-frequency projections is empirically optimal transport (OT), resulting in significantly lower trajectory curvature [24]. “Isokinetic Flow Matching” introduces a Jacobian-free regularizer that penalizes the material derivative of the velocity field, suppressing pathwise acceleration and effectively straightening flows without requiring explicit OT calculations [25]. Additionally, “Constant Acceleration Flow” (CAF) enhances expressiveness by integrating a constant acceleration term into the ODE, allowing for closed-form solutions that enable precise one-step sampling [26].

2.1.3 Variational and Multi-Modal Flow Formulations While straight trajectories offer speed, they can struggle with multi-modal distributions where a single deterministic path may not suffice. To address this, variational formulations have been introduced. “Variational Rectified Flow Matching” (V-RFM) introduces a latent variable to condition the velocity vector field, enabling the modeling of multi-modal flow directions at identical space-time locations [27]. This approach captures inherent multi-modality in source-target couplings by modeling velocity fields as Gaussian mixtures.

Similarly, “Gaussian Mixture Flow Matching” (GMFlow) replaces single Gaussian velocity predictions with dynamic Gaussian mixture parameters, deriving analytic reverse transition distributions for error-free single-step sampling [28]. These methods demonstrate that while straight flows are efficient, introducing stochasticity or latent conditioning is crucial for capturing complex data manifolds, particularly in few-step generation scenarios.

2.2 Architectural Strategies for Conditional Modeling

Effective controllability often requires architectural modifications to inject condition signals without compromising the generative prior. This section reviews adapter-based methods, condition-aware reparameterization, and unified multi-condition frameworks.

2.2.1 Adapter-Based and Parameter-Efficient Control Adapter networks have become the de facto standard for injecting control signals into frozen pre-trained models. “UniCon” implements a unidirectional information flow from a frozen diffusion model to a trainable adapter, eliminating backward gradients through the backbone and reducing training VRAM by 50% compared to

ControlNet [13]. “T2I-Adapter” employs lightweight convolutional networks to extract multi-scale condition features (e.g., sketches, depth) and aligns them with internal UNet representations via element-wise addition, unlocking implicit structural knowledge in frozen models [12].

For more complex interactions, “ControlNet-XS” rethinks control as a feedback-control system, replacing sparse feedback loops with bidirectional, high-frequency connections between generative and controlling encoders. This allows smaller controllers to focus solely on guidance rather than generation, outperforming standard ControlNet in fidelity [29]. In the flow matching domain, “Att-Adapter” integrates decoupled cross-attention modules with a Conditional Variational Autoencoder (CVAE) regularization to prevent identity saturation in multi-attribute control, enabling robust domain-specific adaptation [30].

2.2.2 Condition-Aware Reparameterization and Source Learning Beyond adapters, modifying the source or target distributions themselves can enhance controllability. “Condition-Aware Reparameterization for Flow Matching” (CAR-Flow) introduces lightweight, learnable shift maps to reparameterize source and target distributions, aligning class-conditioned manifolds in latent space to shorten transport distances and prevent mode collapse [31]. Similarly, “Better Source, Better Flow” (CSFM) replaces the fixed Gaussian source with a learnable conditional Gaussian, utilizing variance-only regularization to prevent collapse while allowing mean adaptation to the target [32].

“Learned Distribution-guided Flow Matching” (LeDiFlow) takes this further by replacing static priors with a learned distribution predicted by an auxiliary VAE-like model. This simplifies the transport map, accelerating inference by 3.75x while maintaining quality [33]. These approaches highlight that optimizing the boundary conditions of the flow (source and target) is as critical as optimizing the velocity field itself for efficient conditional generation.

2.2.3 Unified Multi-Condition and Compositional Frameworks Handling multiple, potentially conflicting conditions remains a challenge. “AnyControl” introduces a Multi-Control Encoder utilizing learnable query tokens to unify diverse spatial conditions and textual prompts, alternating between fusion and alignment blocks to dynamically learn composition weights [17]. “UniGen” employs Condition Modulated Experts (CoMoE) to aggregate semantically similar patch features into dedicated experts, reducing redundancy across 12 condition types while maintaining distinct conditional modeling [34].

For layout-controlled generation, “HiCo” (Hierarchical Controllable) employs weight-shared multi-branch networks to independently model background and foreground instances, achieving superior spatial disentanglement compared to single-branch attention mechanisms [18]. “DC-ControlNet” decouples global control into hierarchical elements via Intra-Element and Inter-Element Controllers, separating content attribute control from spatial ordering to eliminate melting artifacts in overlapping regions [35]. These architectures demonstrate a shift towards modular, hierarchical designs that can handle the complexity of real-world multi-modal inputs.

2.3 Training-Free and Inference-Time Control

A significant advantage of flow-based and diffusion models is the ability to exert control at inference time without additional training. This includes posterior sampling, vector field steering, and optimization-based guidance.

2.3.1 Posterior Sampling and Inverse Problems Solving inverse problems (e.g., super-resolution, inpainting) often requires sampling from a posterior distribution $p(x|y)$. “Flow Matching Posterior Sampling” (FMPS) introduces a condition-related correction term into the velocity field ODE, reformulating the velocity function to expose an implicit score function recoverable via Bayes’ rule [9]. This enables training-free conditional generation for pre-trained unconditional flow models.

Similarly, “Flow-Driven Posterior Sampling” (FlowDPS) decomposes the flow ODE into clean image and noise components via a flow-version of Tweedie’s formula, integrating likelihood gradients solely into the clean estimate term to balance data consistency and perceptual quality [36]. For linear inverse problems, “Training-Free Linear Image Inverses via Flows” adapts gradient correction methods to flow models by modifying the vector field with likelihood gradients scaled by path-specific weights, leveraging the straightness of Conditional Optimal Transport paths to stabilize corrections [37].

2.3.2 Vector Field Steering and Guidance Steering the generative trajectory directly in the vector field space offers precise control. “FlowChef” steers rectified flow trajectories by approximating gradients directly in the vector field, bypassing backpropagation through ODE solvers. This leverages the straight-line property of Rectified Flow Models to ensure exponential convergence of error dynamics [10]. “Steering Rectified Flow Models in the Vector Field” further demonstrates that gradient skipping via vector field linearity assumptions enables a unified, inversion-free framework for editing and inverse problems [10].

Guidance mechanisms have also been refined. “Probability-Conserving Flow Guidance” (AdaMaG) decomposes the guidance field into score-parallel and orthogonal components, attenuating the score-parallel flux to constrain updates to iso-density surfaces, thereby preserving the data manifold [38]. “Momentum Guidance” maintains an exponential moving average of past velocities along the ODE trajectory to construct a smoother reference signal, achieving guidance effects with zero additional network function evaluations [39]. “Normalized Attention Guidance” (NAG) performs extrapolation in attention feature space with L1-norm normalization, preserving semantic coherence when conditional branches diverge [40].

2.3.3 Optimization-Based and Energy-Guided Sampling Optimization-based approaches treat generation as an energy minimization problem. “MultiDiffusion” fuses multiple independent diffusion paths via least-squares reconciliation at each denoising step, adapting fixed models to arbitrary aspect ratios and spatial layouts without fine-tuning [41]. “TraDiffusion” utilizes a distance-aware energy function to optimize latent variables during initial diffusion steps, effectively guiding cross-attention maps to follow user-defined trajectories [42].

Energy-based guidance has also been extended to flow models. “Feynman-Kac-Flow” injects stochasticity into deterministic CFM ODEs via score-based correction terms, deriving Feynman-Kac potentials to steer generation towards energy-tilted posteriors while preserving marginals [43]. “Generative Visual Prompt” defines control as latent-space Energy-Based Models (EBMs) approximated by Invertible Neural Networks (INNs), enabling unified distributional control for attributes like de-biasing and style transfer [44].

2.4 Spatial, Semantic, and Attribute Control

Precise control over spatial layout, semantic content, and fine-grained attributes is essential for applications like virtual try-on, medical imaging, and artistic creation.

2.4.1 Spatial and Layout Control Spatial control methods often rely on modifying attention mechanisms or injecting layout embeddings. “LayoutDiffusion” transforms image-layout fusion into a unified form by constructing structural image patches with region info and employing an Object-aware Cross Attention mechanism [45]. “Masked-Attention Guidance” (MAG) optimizes noise maps via gradient descent on a masked-attention loss, maximizing cross-attention values between target words and mask regions while minimizing attention outside masks [46].

For more complex 3D layouts, “Build-A-Scene” (BAS) reformulates text-to-image generation as a sequential multi-stage process, integrating new objects via Dynamic Self-Attention (DSA) to preserve background consistency while adding foreground elements [47]. “LocRef-Diffusion” integrates Appearance-Net and Layout-net into a frozen UNet, using region-masked cross-attention to prevent spatial leakage among multiple instances [48]. In the flow domain, “LayoutFlow” models layout generation as learning a time-dependent vector field via Conditional Flow Matching, producing smooth, directed paths from noise to layout data [49].

2.4.2 Semantic and Identity Preservation Preserving identity while modifying attributes is a key challenge in person image synthesis and editing. “Leffa” (Learning Flow Fields in Attention) constructs flow fields by multiplying averaged attention maps with normalized coordinate maps to warp reference images onto target spaces, explicitly supervising attention maps to mitigate texture distortion [50]. “DiscoHuman” employs a stage-by-stage decomposition using a DiT-based architecture to synthesize human images from variable semantic conditions (face, clothing), avoiding factor entanglement caused by inconsistent data forms [51].

For identity preservation in subject-driven generation, “INSTRUCT IDENTITY” concatenates noise-free reference latents with noisy target latents within a FLUX.1 architecture, enforcing a spatially masked Conditional Flow Matching loss to prevent identity leakage [52]. “PSTF-AttControl” employs a Triplet-Decoupled Cross-Attention module to independently process text, identity, and attribute embeddings, mapping attributes to StyleGAN2 W+ space for fine-grained editing without retraining [53].

2.4.3 Fine-Grained Attribute and Aesthetic Control Controlling specific attributes like lighting, age, or aesthetic quality requires disentangled representations. “AttriCtrl” quantifies aesthetic attributes via direct metrics or CLIP-based similarity, mapping normalized scalar values to learnable token sequences via sinusoidal encoding for precise, disentangled control [16]. “CompSlider” replaces the CLIP image encoder with a DiT model predicting pure image conditions from text tokens and embedded slider values, utilizing a disentanglement loss to break spurious correlations between attributes [54].

Lighting control is addressed by “DiLightNet,” which utilizes radiance hints (renderings of coarse geometry) modulated by encoded provisional images to guide diffusion sampling, demonstrating that coarse approximations suffice for effective relighting [55]. “Bokeh Diffusion” injects physical defocus blur parameters via lightweight cross-attention adapters, employing grounded self-attention to anchor scene structure while varying blur levels [56]. In the flow domain, “BokehFlow” models bokeh generation as direct latent space transport from all-in-focus to bokeh distributions, eliminating Gaussian noise initialization to reduce trajectory complexity [57].

2.5 Acceleration and Distillation Techniques

While flow models are inherently faster than diffusion models, further acceleration is achieved through distillation, one-step generation, and optimized solvers.

2.5.1 Distillation and One-Step Generation Distillation techniques aim to train student models that mimic teacher trajectories in fewer steps. “Flow Generator Matching” (FGM) minimizes the L2 distance between an implicit student vector field and a pre-trained teacher flow, deriving a tractable objective via the Flow Product Identity to enable one-step distillation without explicit density estimation [58]. “Trajectory Consistency Distillation” (TCD) replaces standard skip-connection parameterization with a Trajectory Consistency Function utilizing semi-linear structures, broadening consistency boundary conditions to arbitrary timesteps [59].

“PI-FLOW” (Policy-based Flow) decouples network evaluation steps from ODE integration sub-steps, predicting a network-free policy function that generates dynamic flow velocities for dense intermediate substeps, enabling few-step generation with high fidelity [60]. “Transition Flow Matching” (TFM) directly models global transition flows rather than local velocities, deriving a Transition Flow Identity that enables single-step generation from scratch [61].

2.5.2 Optimized Solvers and Sampling Schemes Improving the ODE solver itself can yield significant speedups. “Bespoke Solvers” construct custom ODE solvers via learnable scale-time transformations applied to base solvers, optimizing time reparametrization to minimize global truncation error specific to the model’s trajectory characteristics [62]. “Bellman Optimal Step-size Straightening” (BOSS) formulates sampling schedule optimization as a network flow problem solved via dynamic programming, minimizing accumulated Euler truncation error under fixed NFE budgets [63].

“Bi-Anchor Interpolation Solver” (BA-solver) integrates a lightweight SideNet with a frozen Flow Matching backbone to predict velocity deviations conditioned on bidirectional offsets, matching 100+ NFE Euler solver quality at only 15 NFEs [64]. “Warm-Start Flow Matching” (WS-FM) replaces pure noise initialization with draft samples from lightweight models, shifting the flow matching start time to $t_0 > 0$ to guarantee a speed-up factor of $1/(1 - t_0)$ [65].

2.6 Domain-Specific Applications

The versatility of diffusion and flow models has led to their adoption in specialized domains requiring high precision and controllability.

2.6.1 Medical Image Synthesis In medical imaging, controllability is critical for data augmentation and diagnosis. “MedSyn2” employs a modified Diffusion Transformer with compound prompting, combining semantic mask descriptions with radiology reports to generate 3D CT scans with flexible control [66]. “FlowLet” integrates Flow Matching within an invertible 3D Haar wavelet domain for artifact-free synthesis of brain MRI, reducing memory usage by 8x while preserving anatomical details [67]. “CrackSegFlow” employs a two-stage Flow Matching pipeline for crack segmentation, utilizing topology-preserving SPADE normalization to ensure sub-pixel crack continuity [68].

2.6.2 Remote Sensing and Geospatial Data Remote sensing applications often require conditioning on geographic metadata and terrain. “GeoDiT” implements a DiT backbone with Flow

Matching and an Adaptive Local Attention module, enabling point-based conditioning for satellite image synthesis without dense pixel-level maps [69]. “D2-CDIG” integrates dual ControlNet branches to decouple terrain (DEM) and atmospheric (cloud-fog) generation processes, preventing interference between geometric constraints and stochastic atmospheric effects [70]. “TODSynth” utilizes a Multimodal Diffusion Transformer with Control-Rectify Flow Matching to inject semantic loss gradients for precise spatial control in semantic segmentation tasks [71].

2.6.3 Video and Motion Control Extending control to video involves managing temporal consistency and camera motion. “MotionCanvas” integrates a Motion Design Module to convert 3D scene-space intents into robust 2D screen-space conditioning signals, enabling joint control of camera and object motion [72]. “FloVD” employs a two-stage pipeline where optical flow represents 3D correlations across viewpoints, allowing detailed camera control without explicit camera parameters [73]. “DEMO” (Disentangled Motion Latent Flow Matching) constructs a fine-grained motion latent space via contrastive learning, applying optimal-transport flow matching to map audio conditions to independent lip, pose, and eye movements for talking head synthesis [74].

2.6.4 Robotics and Action Generation In robotics, generative models are used for action planning and visuomotor control. “NinA” (Normalizing Flows in Action) replaces diffusion action decoders with invertible Normalizing Flows conditioned on VLM embeddings, enabling one-shot action sampling with 10x faster inference [75]. “GENIMA” reframes joint-action generation as an image-to-image translation task, rendering future joint positions as colored spheres onto current observations to leverage internet-scale diffusion priors for policy generation [76]. “Trajectory Conditional Flow Matching” (T-CFM) applies flow matching to conditional trajectory prediction, utilizing straight probability flows for real-time planning in robotics [77].

3. Challenges and Outlook

Despite significant advancements, several challenges remain in the development and deployment of controllable diffusion and flow-based models.

3.1 Evaluation Bottlenecks and Metric Limitations

Current evaluation metrics often fail to capture the nuances of controllability. Standard metrics like FID and IS measure overall distribution quality but do not assess adherence to specific conditions (e.g., layout accuracy, attribute intensity). While some works propose task-specific metrics (e.g., IoU for segmentation, LPIPS for editing), there is a lack of unified benchmarks for multi-condition control [29, 78]. “FLD+” proposes a flow-based likelihood distance metric that is more data-efficient than FID, but its dependence on pre-trained backbone features may introduce domain bias [78]. Furthermore, human evaluation remains the gold standard for assessing aesthetic and semantic alignment, yet it is costly and subjective [79, 80]. Future research must develop comprehensive, automated metrics that quantify both generative quality and control precision.

3.2 Safety, Bias, and Alignment

As generative models become more powerful, risks related to bias, misinformation, and malicious use increase. “DiffInf” highlights the issue of noisy labels and proposes influence-guided diffusion to repair training data, yet broader strategies for debiasing are needed [81]. “Generative Visual Prompt” demonstrates the potential for de-biasing via latent-space EBMs, but output distribution

support is restricted by the pre-trained model’s coverage [44]. Alignment with human values is also critical; methods like “Flow-GRPO” and “SuperFlow” integrate reinforcement learning to align models with reward signals, but they remain vulnerable to reward hacking and credit assignment issues [82, 83]. Ensuring robust safety guardrails without compromising generative diversity is an open problem.

3.3 Computational Efficiency and Deployment

While flow models offer faster sampling than diffusion models, high-resolution generation and complex conditioning still incur significant computational costs. Methods like “ControlNet-XS” and “RelaCtrl” aim to reduce parameter overhead, but inference latency is often dominated by the main backbone network [29, 84]. Distillation techniques (e.g., FGM, TCD) enable one-step generation but require expensive teacher-student training pipelines [58, 59]. Deploying these models on edge devices remains challenging due to memory constraints and the need for specialized hardware accelerators [3]. Future work should focus on developing lightweight architectures, efficient distillation protocols, and hardware-aware optimization strategies.

3.4 Theoretical Gaps and Generalization

Theoretical understanding of flow-based models is still evolving. While connections to optimal transport and stochastic control are established, the behavior of learned velocity fields in high-dimensional spaces is not fully characterized. “On the Closed-Form of Flow Matching” suggests that deterministic target regression improves generalization, yet the precise mechanisms remain unclear [85]. Additionally, generalization to out-of-domain scenarios (e.g., novel objects, unseen layouts) remains a limitation for many conditional models [86, 51]. Bridging the gap between theoretical guarantees and empirical performance, especially in few-shot and zero-shot settings, is a critical direction for future research.

4. Conclusion

The field of controllable image generation has witnessed a transformative shift with the rise of diffusion and flow-based models. By moving from stochastic SDEs to deterministic ODEs, flow matching and rectified flow approaches have unlocked new levels of efficiency and controllability. This survey has synthesized a vast body of literature, highlighting key innovations in trajectory straightening, architectural conditioning, inference-time steering, and domain-specific applications. From theoretical unifications of probability transport to practical frameworks for medical imaging and robotics, these models demonstrate unprecedented flexibility and power.

However, challenges persist in evaluation, safety, efficiency, and generalization. Addressing these issues will require interdisciplinary efforts combining advances in optimization theory, computer vision, and ethical AI. As the field continues to evolve, we anticipate further convergence of diffusion and flow paradigms, leading to generative systems that are not only high-fidelity and fast but also deeply controllable, safe, and accessible for diverse real-world applications. The journey from noise to structured data is becoming increasingly direct, promising a future where generative AI serves as a precise and reliable tool for creativity and discovery.

References

- [1] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma. (2021). Score-Based Generative Mod-

eling through Stochastic Differential Equations. arXiv:2011.13456.

[2] Yang Song, Stefano Ermon. (2020). Improved Techniques for Training Score-Based Generative Models. arXiv:2006.09011.

[3] Srishti Gupta, Yashasvee Taiwade. (2024). Efficiency vs. Fidelity: A Comparative Analysis of Diffusion Probabilistic Models and Flow Matching on Low-Resource Hardware. arXiv:2511.19379.

[4] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu. (2023). FLOW MATCHING FOR GENERATIVE MODELING. arXiv:2210.02747.

[5] Peter Holderrieth, Ezra Erives. (2025). An Introduction to Flow Matching and Diffusion Models. arXiv:2506.02070.

[6] Shiv Shankar, Tomas Geffner. (2024). LEARNING STRAIGHT FLOWS BY LEARNING CURVED INTERPOLANTS. arXiv:2503.20719.

[7] Onur Tasar, Clement Chadebec, Benjamin Aubin. (2024). Controllable Shadow Generation with Single-Step Diffusion Models from Synthetic Data. arXiv:2412.11972.

[8] Shin seong Kim, Mingi Kwon, Jaeseok Jeong. (2024). Balanced Conic Rectified Flow. arXiv:2510.25229.

[9] Kaiyu Song, Hanjiang Lai, Yan Pan. (2024). FLOW MATCHING POSTERIOR SAMPLING: A TRAINING-FREE CONDITIONAL GENERATION FOR FLOW MATCHING. arXiv:2411.07625.

[10] Maitreya Patel, Song Wen, Dimitris N. Metaxas. (2024). Steering Rectified Flow Models in the Vector Field for Controlled Image Generation. arXiv:2412.00100.

[11] Pu Cao, Feng Zhou, Qing Song. (2024). Controllable Generation with Text-to-Image Diffusion Models: A Survey. arXiv:2403.04279.

[12] Chong Mou, Xintao Wang, Liangbin Xie. (2024). T2I-Adapter: Learning Adapters to Dig out More Controllable Ability for Text-to-Image Diffusion Models. arXiv:2302.08453.

[13] Fanghua Yu, Jinjin Gu, Jinfan Hu. (2024). UNICON: UNIDIRECTIONAL INFORMATION FLOW FOR EFFECTIVE CONTROL OF LARGE-SCALE DIFFUSION MODELS. arXiv:2503.17221.

[14] Pedro M. P. Curvo, Maksim Zhdanov, Floor Eijkelboom. (2025). Follow the Mean: Reference-Guided Flow Matching. arXiv:2605.10302.

[15] Litu Rout, Yujia Chen, Nataniel Ruiz. (2024). SEMANTIC IMAGE INVERSION AND EDITING USING RECTIFIED STOCHASTIC DIFFERENTIAL EQUATIONS. arXiv:2410.10792.

[16] Die Chen, Zhongjie Duan, Zhiwen Li. (2025). AttriCtrl: Fine-Grained Control of Aesthetic Attribute Intensity in Diffusion Models. arXiv:2508.02151.

[17] Yanan Sun, Yanchen Liu, Yinhao Tang. (2024). AnyControl: Create Your Artwork with Versatile Control on Text-to-Image Generation. arXiv:2406.18958.

[18] Bo Cheng, Yuhang Ma, Liebucha Wu. (2024). HiCo: Hierarchical Controllable Diffusion Model for Layout-to-image Generation. arXiv:2410.14324.

[19] Aditya Ranganath, Mukesh Singhal. (2025). A Unified Measure-Theoretic View of Diffusion, Score-Based, and Flow Matching Generative Models. arXiv:2605.06829.

- [20] Kaizhen Zhu, Mokai Pan, Zhechuan Yu. (2025). DIFFUSION BRIDGE OR FLOW MATCHING? A UNIFYING FRAMEWORK AND COMPARATIVE ANALYSIS. arXiv:2509.24531.
- [21] Yaron Lipman, Marton Havasi, Peter Holderrieth. (2024). Flow Matching Guide and Code. arXiv:2412.06264.
- [22] Shujian Zhang, Lemeng Wu, Chengyue Gong. (2023). Language Rectified Flow: Advancing Diffusion Language Generation with Probabilistic Flows. arXiv:2403.16995.
- [23] Jongmin Yoon, Juho Lee. (2024). Sequential Flow Straightening for Generative Modeling. arXiv:2402.06461.
- [24] Shimon Malnick, Matan Rusanovsky, Ohad Fried. (2024). Designing Optimal Transport Flows. arXiv:2606.04092.
- [25] Tauhid Khan. (2024). Isokinetic Flow Matching for Pathwise Straightening of Generative Flows. arXiv:2604.04491.
- [26] Dogyun Park, Sojin Lee, Sihyeon Kim. (2024). Constant Acceleration Flow. arXiv:2411.00322.
- [27] Pengsheng Guo, Alexander G. Schwing. (2024). Variational Rectified Flow Matching. arXiv:2502.09616.
- [28] Hansheng Chen, Kai Zhang, Hao Tan. (2024). Gaussian Mixture Flow Matching Models. arXiv:2504.05304.
- [29] Denis Zavadski, Johann-Friedrich Feiden, Carsten Rother. (2024). ControlNet-XS: Rethinking the Control of Text-to-Image Diffusion Models as Feedback-Control Systems. arXiv:2312.06573.
- [30] Wonwoong Cho, Yan-Ying Chen, Matthew Klenk. (2024). Att-Adapter: A Robust and Precise Domain-Specific Multi-Attributes T2I Diffusion Adapter via Conditional Variational Autoencoder. arXiv:2503.11937.
- [31] Chen Chen, Pengsheng Guo, Liangchen Song. (2025). CAR-Flow: Condition-Aware Reparameterization Aligns Source and Target for Better Flow Matching. arXiv:2509.19300.
- [32] Junwan Kim, Jiho Park, Seonghu Jeon. (2025). Better Source, Better Flow: Learning Condition-Dependent Source Distribution for Flow Matching. arXiv:2602.05951.
- [33] Pascal Zwick, Nils Friederich, Maximilian Beichter. (2024). LeDiFlow: Learned Distribution-guided Flow Matching to Accelerate Image Generation. arXiv:2505.20723.
- [34] Guoqing Zhang, Xingtong Ge, Lu Shi. (2025). Condition Weaving Meets Expert Modulation: Towards Universal and Controllable Image Generation. arXiv:2508.17364.
- [35] Hongji Yang, Wencheng Han, Yucheng Zhou. (2024). DC-ControlNet: Decoupling Inter- and Intra-Element Conditions in Image Generation with Diffusion Models. arXiv:2502.14779.
- [36] Jeongsol Kim, Bryan Sangwoo Kim, Jong Chul Ye. (2024). FlowDPS: Flow-Driven Posterior Sampling for Inverse Problems. arXiv:2503.08136.
- [37] Ashwini Pokle, Matthew J. Muckley, Ricky T. Q. Chen. (2024). TRAINING-FREE LINEAR IMAGE INVERSES VIA FLOWS. arXiv:2310.04432.
- [38] Parsa Esmati, Junha Hyung, Amirhossein Dadashzadeh. (2025). Probability-Conserving Flow Guidance. arXiv:2605.20079.

- [39] Runlong Liao, Jian Yu, Baiyu Su. (2024). Momentum Guidance: Plug-and-Play Guidance for Flow Models. arXiv:2602.20360.
- [40] Dar-Yen Chen, Hmrishav Bandyopadhyay, Kai Zou. (2024). Normalized Attention Guidance: Universal Negative Guidance for Diffusion Models. arXiv:2505.21179.
- [41] Omer Bar-Tal, Lior Yariv, Yaron Lipman. (2023). MultiDiffusion: Fusing Diffusion Paths for Controlled Image Generation. arXiv:2302.08113.
- [42] Mingrui Wu, Oucheng Huang, Jiayi Ji. (2024). TraDiffusion: Trajectory-Based Training-Free Image Generation. arXiv:2408.09739.
- [43] Konstantin Mark, Leonard Galustian, Maximilian P.-P. Kovar. (2025). Feynman-Kac-Flow: Inference Steering of Conditional Flow Matching to an Energy-Tilted Posterior. arXiv:2509.01543.
- [44] Chen Henry Wu, Saman Motamed, Shaunak Srivastava. (2024). Generative Visual Prompt: Unifying Distributional Control of Pre-Trained Generative Models. arXiv:2209.06970.
- [45] Guangcong Zheng, Xianpan Zhou, Xuewei Li. (2023). LayoutDiffusion: Controllable Diffusion Model for Layout-to-image Generation. arXiv:2303.17189.
- [46] Yuki Endo. (2024). Masked-Attention Diffusion Guidance for Spatially Controlling Text-to-Image Generation. arXiv:2308.06027.
- [47] Abdelrahman Eldesokey, Peter Wonka. (2024). BUILD-A-SCENE: INTERACTIVE 3D LAYOUT CONTROL FOR DIFFUSION-BASED IMAGE GENERATION. arXiv:2408.14819.
- [48] Fan Deng, Xinyang Yu, Xiangjun Huang. (2024). LocRef-Diffusion: Tuning-Free Layout and Appearance-Guided Generation. arXiv:2411.15252.
- [49] Julian Jorge Andrade Guerreiro, Naoto Inoue, Kento Masui. (2024). LayoutFlow: Flow Matching for Layout Generation. arXiv:2403.18187.
- [50] Zijian Zhou, Shikun Liu, Xiao Han. (2024). Learning Flow Fields in Attention for Controllable Person Image Generation. arXiv:2412.08486.
- [51] Zhengwentai Sun, Chenghong Li, Hongjie Liao. (2025). Exploring Disentangled and Controllable Human Image Synthesis: From End-to-End to Stage-by-Stage. arXiv:2503.19486.
- [52] Aditi Singhania, Arushi Jain, Krutik Malani. (2025). Taming Identity Consistency and Prompt Diversity in Diffusion Models via Latent Concatenation and Masked Conditional Flow Matching. arXiv:2511.08061.
- [53] Xiang Liu, Zhaoxiang Liu, Huan Hu. (2024). PSTF-AttControl: Per-Subject-Tuning-Free Personalized Image Generation with Controllable Face Attributes. arXiv:2510.25084.
- [54] Zixin Zhu, Kevin Duarte, Mamshad Nayeem Rizve. (2024). CompSlider: Compositional Slider for Disentangled Multiple-Attribute Image Generation. arXiv:2509.01028.
- [55] Chong Zeng, Yue Dong, Pieter Peers. (2024). DiLightNet: Fine-grained Lighting Control for Diffusion-based Image Generation. arXiv:2402.11929.
- [56] Armando Fortes, Tianyi Wei, Shangchen Zhou. (2024). Bokeh Diffusion: Defocus Blur Control in Text-to-Image Diffusion Models. arXiv:2503.08434.
- [57] Yachuan Huang, Xianrui Luo, Qiwen Wang. (2025). BokehFlow: Depth-Free Controllable Bokeh Rendering via Flow Matching. arXiv:2511.15066.

- [58] Zemin Huang, Zhengyang Geng, Weijian Luo. (2024). Flow Generator Matching. arXiv:2410.19310.
- [59] Jianbin Zheng, Minghui Hu, Zhongyi Fan. (2024). TRAJECTORY CONSISTENCY DISTILLATION: Improved Latent Consistency Distillation by Semi-Linear Consistency Function with Trajectory Mapping. arXiv:2402.19159.
- [60] Hansheng Chen, Kai Zhang, Hao Tan. (2025). PI-FLOW: POLICY-BASED FEW-STEP GENERATION VIA IMITATION DISTILLATION. arXiv:2510.14974.
- [61] Chenrui Ma. (2024). Transition Flow Matching. arXiv:2603.15689.
- [62] N. Shaul, J. Perez, R. T. Q. Chen. (2024). Bespoke Solvers for Generative Flow Models. arXiv:2310.19075.
- [63] Bao Nguyen, Binh Nguyen, Viet Anh Nguyen. (2024). BELLMAN OPTIMAL STEPSIZE STRAIGHTENING OF FLOW-MATCHING MODELS. arXiv:2312.16414.
- [64] Yusheng Mu, Qingting Xu, Author Three. (2025). Bi-Anchor Interpolation Solver for Accelerating Generative Modeling. arXiv:2601.21542.
- [65] Minyoung Kim. (2025). Warm-Start Flow Matching for Guaranteed Fast Text/Image Generation. arXiv:2603.19360.
- [66] Weicheng Dai, Chenyu Wang, Andy Li. (2024). MedSyn2: Flexible Control of 3D CT Generation via Text and Semantically-Defined Segmentation Prompts. arXiv:2606.00967.
- [67] Danilo Danese, Angela Lombardi, Matteo Attimonelli. (2025). FLOWLET: CONDITIONAL 3D BRAIN MRI SYNTHESIS USING WAVELET FLOW MATCHING. arXiv:2601.05212.
- [68] Babak Asad, Peiyang Wu, Mani Golparvar-Fard. (2024). CrackSegFlow: Controllable Flow Matching Synthesis for Generalizable Crack Segmentation with a 50K Image–Mask Benchmark. arXiv:2601.03637.
- [69] Srikanth Sastry, Dan Cher, Brian Wei. (2024). GeoDiT: Point-Conditioned Diffusion Transformer for Satellite Image Synthesis. arXiv:2603.02172.
- [70] Zuopeng Zhao, Ying Liu, Kanyaphakphachorn Pharksuwan. (2024). D2-CDIG: Controlled Diffusion Remote Sensing Image Generation with Dual Priors of DEM and Cloud-Fog. arXiv:2605.14326.
- [71] Yunkai Yang, Yudong Zhang, Kunquan Zhang. (2024). Task-Oriented Data Synthesis and Control-Rectify Sampling for Remote Sensing Semantic Segmentation. arXiv:2512.16740.
- [72] Jinbo Xing, Long Mai, Cusuh Ham. (2025). MotionCanvas: Cinematic Shot Design with Controllable Image-to-Video Generation. arXiv:2502.04299.
- [73] Wonjoon Jin, Qi Dai, Chong Luo. (2024). FloVD: Optical Flow Meets Video Diffusion Model for Enhanced Camera-Controlled Video Synthesis. arXiv:2502.08244.
- [74] Peiyin Chen, Zhuowei Yang, Hui Feng. (2024). DEMO: Disentangled Motion Latent Flow Matching for Fine-Grained Controllable Talking Portrait Synthesis. arXiv:2510.10650.
- [75] Denis Tarasov, Alexander Nikulin, Ilya Zisman. (2025). NinA: Normalizing Flows in Action. Training VLA Models with Normalizing Flows. arXiv:2508.16845.

- [76] Mohit Shridhar, Yat Long Lo, Stephen James. (2024). Generative Image as Action Models. arXiv:2407.07875.
- [77] Sean Ye, Matthew C. Gombolay. (2024). Efficient Trajectory Forecasting and Generation with Conditional Flow Matching. arXiv:2403.10809.
- [78] Pranav Jeevan, Neeraj Nixon, Amit Sethi. (2024). FLD+: Data-Efficient Evaluation Metric for Generative Models. arXiv:2411.15584.
- [79] Junzhi Ning, Cheng Tang, Kaijing Zhou. (2024). RetinaLogos: Fine-Grained Synthesis of High-Resolution Retinal Images Through Captions. arXiv:2505.12887.
- [80] Yifan Xie, Jingge Wang, Tao Feng. (2024). CCIS-DIFF: A GENERATIVE MODEL WITH STABLE DIFFUSION PRIOR FOR CONTROLLED COLONOSCOPY IMAGE SYNTHESIS. arXiv:2411.12198.
- [81] Basudha Pala, Rama Chellappa. (2025). DiffInf: Influence-Guided Diffusion for Supervision Alignment in Facial Attribute Learning. arXiv:2603.06399.
- [82] Jie Liu, Gongye Liu, Jiajun Liang. (2024). Flow-GRPO: Training Flow Matching Models via Online RL. arXiv:2505.05470.
- [83] Kaijie Chen, Zhiyang Xu, Ying Shen. (2025). SuperFlow: Training Flow Matching Models with RL on the Fly. arXiv:2512.17951.
- [84] Ke Cao, Jing Wang, Ao Ma. (2025). RelaCtrl: Relevance-Guided Efficient Control for Diffusion Transformers. arXiv:2502.14377.
- [85] Quentin Bertrand, Anne Gagneux, Mathurin Massias. (2025). On the Closed-Form of Flow Matching: Generalization Does Not Arise from Target Stochasticity. arXiv:2506.03719.
- [86] Yuhang Qian, Haiyan Chen, Wentong Li. (2025). Text-guided Controllable Diffusion for Realistic Camouflage Images Generation. arXiv:2511.20218.